

PURPLE TOUCH v4

Filtering, Scanning, Lifecycle, and Windows Setup Guide

A visual implementation reference for Monthly, Weekly, and Daily profiles on branch scanner-audit-and-v4.

Purpose

Explain exactly what data is downloaded, how each filter is evaluated, when a setup enters or leaves monitoring, how alerts and exits are created, and how to run v4 beside an existing v3 installation on Windows.

Candle service

Continuously refreshes prioritized source candles while the monitor is running.

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Setup scanners

Monthly, Weekly, and Daily independently find physical Purple EMA9 touches and strict candidates.

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Persistent setup bucket

Qualified touch lifecycles remain available for repeated Early, Final, and invalidation checks.

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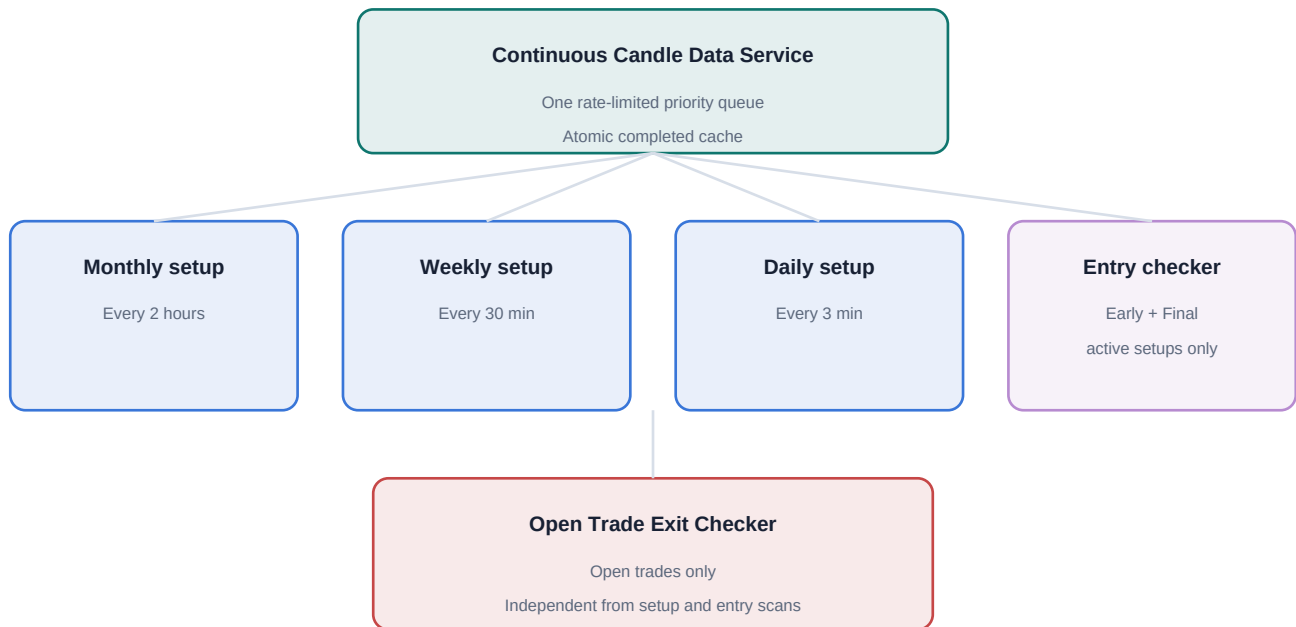
Trade lifecycle

Entry signals create independent trade IDs; only open trades are checked for exits.

Read-only alerting and audit support. The application does not place orders and is not financial advice.

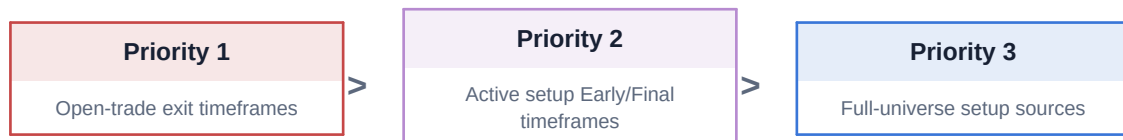
1 Architecture: downloads and analysis are decoupled

The old refresh-then-scan chain has been replaced. Cached-data analysis can run concurrently, but Zerodha downloads remain serialized and rate limited.



Key behavior

Analyze fresh cache now does not download candles. Start candle service & monitor starts the downloader and recurring workers. Scanners never read a partially written CSV and never create alerts from stale data.



One failed symbol or profile is recorded and skipped; it does not terminate the other workers.

2 Timeframe matrix and source files

Profile	Setup touch	Higher confirmation	Early	Final	Exit	Default schedule
Monthly	Monthly	Derived 5-month: open OR close > blue	2 hour	Daily	2 hour	Every 2 hours
Weekly	Weekly	Monthly: open OR close > blue	30 min	2 hour	30 min	Every 30 min
Daily	Daily	Weekly: open OR close > blue	10 min	30 min	10 min	Every 3 min

Why only four source timeframes appear in Candle Data Service

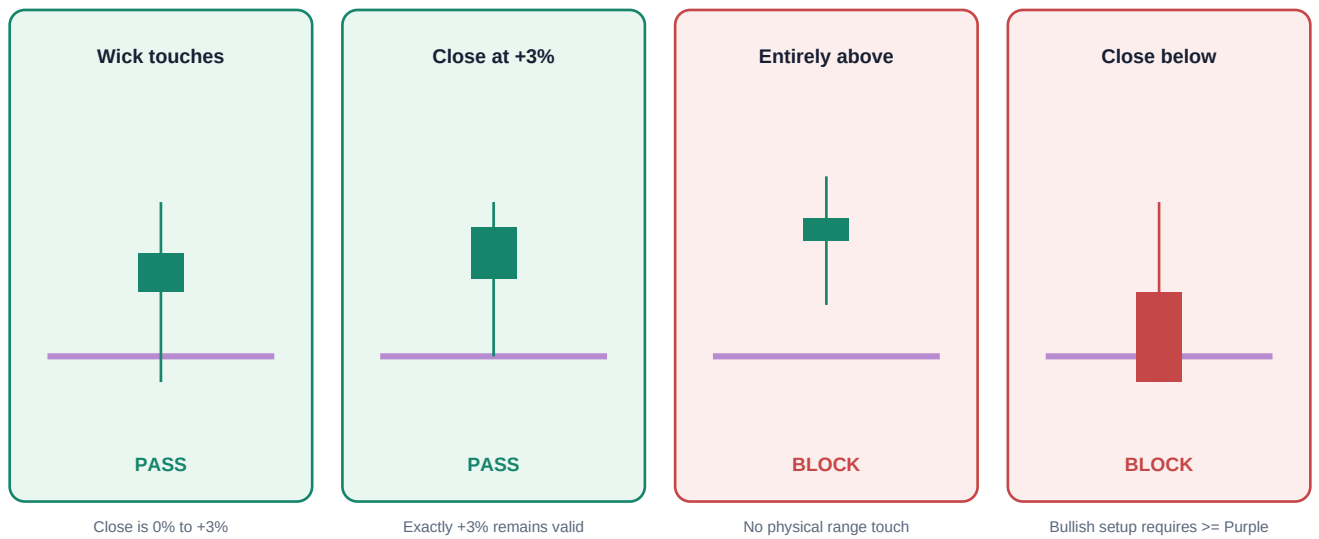
Monthly, Weekly, and derived 5-month candles are built from Daily data. Two-hour candles are built from 1-hour data. Therefore the downloaded sources are Daily, 1 hour, 30 min, and 10 min.

Displayed requirement	Downloaded source	Used by
Monthly / Weekly / Daily / 5-month	Daily	All setup scans and higher confirmation
2 hour	1 hour	Monthly Early/Exit; Weekly Final
30 min	30 min	Weekly Early/Exit; Daily Final
10 min	10 min	Daily Early/Exit

All timestamps stored or displayed by the Purple Touch workflow are converted to Indian Standard Time.

3 Step 1: exact bullish Purple EMA9 touch

The latest setup-timeframe candle must physically contain Purple EMA9 inside its low/high range. The close must be on or above Purple and no more than 3.00% above Purple.



Implemented expression
PASS when $low \leq Purple\ EMA9 \leq high$ AND $Purple\ EMA9 \leq close \leq Purple\ EMA9 \times 1.03$. Exactly +3.00% passes. An entirely-above near touch and any close below Purple are blocked.

Value recorded	Meaning
Touch price	Purple EMA9 value touched by the wick/range
Touch timestamp	Setup candle timestamp in IST
Detected timestamp	When the scanner persisted the setup
Initial distance	Close distance above captured Purple, ranked from 0.00% upward

4 Strict setup candidate filters

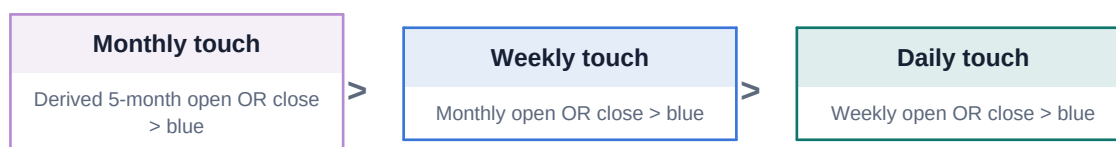
Step 1 rows are audit rows. A stock enters Stocks Waiting for Entry only after all mandatory setup filters pass.

Strict filter	Pass boundary	Role
Physical touch + close zone	Range touches Purple; close is 0.00% to 3.00% above	Mandatory
Black EMA89	Touch-timeframe close strictly > EMA89	Mandatory
Light-green EMA26	Touch-timeframe close >= EMA26	Mandatory
Blue vs Purple	Touch-timeframe blue strictly > Purple EMA9	Mandatory
Approach direction	Prior 2-3 setup candles come from blue area toward Purple	Mandatory
Higher confirmation	Latest mapped candle open OR close strictly > its blue	Mandatory
Market structure	Must not be downtrend	Mandatory
EMA9 vs EMA26	EMA9 > EMA26	Score/context; warning if absent
Yellow vs EMA26	Yellow < EMA26	Score/context; warning if absent
Brown vs EMA26, RSI divergence	Displayed relationship / optional quality	Non-blocking context

No strict candidate

The stock passed the physical Purple touch audit but at least one mandatory setup filter blocked it. It stays visible in Step 1 for explanation, but it is not persisted into the active entry-checker bucket.

5 Higher-timeframe confirmation



Boundary case	Result	Reason
Open above blue; close below	PASS	Open OR close is sufficient
Open below blue; close above	PASS	Open OR close is sufficient
Open equals blue; close below	BLOCK	Comparison is strictly above
Open below; close equals blue	BLOCK	Equality does not pass
Blue unavailable	WAIT/BLOCK	Required candle history is insufficient

Derived 5-month candle

For the Monthly profile, source Monthly candles are grouped into stable five-month calendar buckets. The latest derived bucket supplies open, close, and blue-line confirmation values.

This confirmation belongs to setup qualification. It is separate from Early and Final entry-timeframe rules.

6 Persistent setup lifecycle

Strict setup passes

Create or update one row identified by symbol + profile + touch candle timestamp.

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active_waiting

Entry checker monitors only this persistent bucket using newly closed Early and Final candles.

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Early trigger

Create Early trade; keep checking the same setup for Final.

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Final trigger

Create Final trade; suppress any later Early entry for this setup.

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Discard or complete

Keep the historical setup row for audit; never delete it automatically.

Status	Meaning
active_waiting	Qualified setup is waiting for Early/Final entry rules
early_entry_triggered	Early trade exists; Final monitoring continues
final_entry_triggered	Final trade exists; later Early checks stop
discarded_above_3_percent	Latest entry-check price exceeded fixed captured-Purple limit
discarded_c1_c2_low_break	A Candle 3+ low broke the lower C1/C2 low
completed	Lifecycle no longer needs entry monitoring

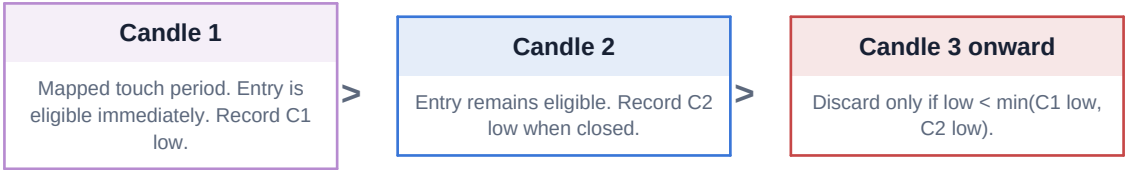
7 Fixed 3% disqualification



Scenario	Setup action	Existing trade action
Price = fixed +3.00% level	Remain active	No change
Price strictly above fixed +3.00%	Discard further Early/Final checks	Keep open; normal Exit monitoring
EMA9 later moves	Do not move threshold	No change
Exceeded during C1, C2, or C3+	Discard setup	No automatic exit

The entry checker uses the latest successfully refreshed price. The stored audit includes actual price, calculated distance, fixed threshold, discard time, and reason.

8 Candle 1 / Candle 2 validity and invalidation



Important

C1 and C2 are not waiting candles. Either can create Early or Final entry if the corresponding entry rules pass. Starting at C3, the same lower C1/C2 low remains the invalidation threshold for every later candle. There is no 10-candle expiry.

Example	Result
C1 low 98; C2 low 97; C3 low 97	Valid: equality does not break below 97
C1 low 98; C2 low 97; C5 low 96.99	Discard: a later low is strictly below 97
C2 not available yet	C1 may trigger entry; later-low invalidation waits for C2

9 Early and Final entry checks

Rule	Early	Final
Required candle	Latest completed Early timeframe	Latest completed Final timeframe
Close strictly above yellow	Mandatory	Mandatory
Yellow strictly above brown VWMA20	Ignored	Mandatory
Blue above Purple on entry timeframe	Ignored / context only	Ignored / context only
RSI bullish divergence	Optional context	Optional context
C1 and C2	Both eligible	Both eligible
Duplicate closed candle	Not reprocessed	Not reprocessed

Active setup + fresh closed candle

Check fixed 3% and C3+ invalidation first.

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Evaluate Final

close > yellow AND yellow > brown. If triggered, suppress later Early.

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Evaluate Early

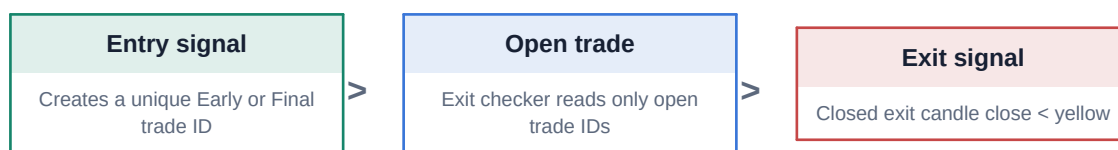
close > yellow. If triggered first, keep setup for Final.

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Persist and alert immediately

Serialize database write, suppress duplicates, optionally send Telegram.

10 Exit alerts and trade history



UI section	What it means
Stocks Waiting for Entry	Persistent qualified setups still being checked, including discard history through filters
Live Scan Alert Summary	Counts and delivery status for the latest analysis
Exit Alerts	Historical exit events created from closed candles below yellow
Trades Triggered: Open & Closed	One lifecycle row per Early/Final trade ID with entry, exit, status, and duration

Independent lifecycles

Early and Final trades for the same setup are separate. Discarding the setup stops new entry checks but does not close an existing trade. Only its configured exit-timeframe closed candle below yellow closes that trade ID.

11 Freshness, scheduling, and UI counts

Freshness state	Meaning	Scanner action
fresh	Successful complete cache is within the source freshness window	May analyze
updating	A queued or active replacement is in progress	Use prior fresh atomic cache only; otherwise defer
stale	A prior success exists but is too old or behind required data	Defer; never alert
missing	No successful refresh exists	Defer; never alert

Why Stale and Missing can be non-zero

The service refreshes a rate-limited queue, not every symbol at once. If the monitor is stopped before completion, completed sources later become stale and unfinished sources remain missing. Analyze fresh cache now does not fill these columns; restart the candle service.

Control	Actual behavior
Start candle service & monitor	Starts continuous downloads, independent schedules, entry checker, and exit checker
Analyze fresh cache now	Immediately analyzes only currently fresh completed sources; may report partial/deferred
Cancel scan	Cooperatively stops an active manual analysis; disabled while idle
Stop	Stops monitor/download workers, clears queued work, preserves setup/trade history

12 Windows: clone v4 into a new folder

Use a new folder so v3 remains untouched. These commands use port 8766; replace it with another free port if needed.

PowerShell commands

```
cd C:\Users\YOUR_NAME\Documents
git clone --branch scanner-audit-and-v4 --single-branch
https://github.com/ajay0589/TradeAnalysisWithAjay.git TradeAnalysis-v4
cd .\TradeAnalysis-v4
Copy-Item .env.example .env
notepad .env
python -m trading_analysis.cli env-check
.\scripts\start_web_ui.ps1 -Port 8766
```

Prerequisite	Check
Git for Windows	git --version
Python 3.11 or newer	python --version
PowerShell execution	If blocked: Set-ExecutionPolicy -Scope Process Bypass
Free port	Get-NetTCPConnection -LocalPort 8766 -ErrorAction SilentlyContinue
Browser	Open http://127.0.0.1:8766

The runtime application uses Python standard-library modules. ReportLab is required only when regenerating this PDF: `python -m pip install reportlab`.

13 Optional: safely carry v3 local data into v4

Git does not copy secrets, downloaded candle caches, logs, reports, or the SQLite database. Stop v3 before copying its database.

Stop v3

Stop its server and confirm no scan/database write is active.

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Copy secrets

Copy v3 .env to v4 .env. Never commit this file.

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Copy cached data

Copy data/raw so v4 does not begin with an empty candle cache.

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Optional history

Copy data/db only if setup, alert, and trade history must continue. v4 migrates schema on startup.

PowerShell example

```
Copy-Item 'C:\path\to\v3\.env' '.\env' -Force
robocopy 'C:\path\to\v3\data\raw' '.\data\raw' /E
robocopy 'C:\path\to\v3\data\db' '.\data\db' /E
```

Do not share one live database folder

v3 and v4 may run simultaneously on different ports only when each folder has its own copied data/db. Do not point both running processes at the same SQLite file.

14 Daily operating checklist

1. Before market, update the Zerodha access token in v4 .env and refresh NSE/NFO instrument masters if needed.
2. At about 9:15-9:20 AM IST, open v4 on port 8766 and click Start candle service & monitor.
3. Expand Candle Data Service. Confirm queue activity, increasing successes, zero/understood failures, and fresh source counts.
4. Check the separate Monthly, Weekly, and Daily profile cards for queued, actual start, completion, duration, analyzed, and deferred counts.
5. Review Stocks Waiting for Entry for captured Purple, fixed +3% level, C1/C2 lows, Early/Final states, and discard reason.
6. Use Analyze fresh cache now only for an immediate audit. It does not download data and can be partial while refresh continues.
7. Keep the PC awake and server running. After 3:30 PM IST, allow active work to finish and then click Stop.

Symptom	Interpretation / action
0 rows in a few seconds	Expand Candle Data Service. If Fresh is 0, start monitor and wait for completed sources.
Many deferred	Normal during queue warm-up; recheck as fresh counts increase.
Failures increase	Inspect Latest failure; verify token, instrument master, network, and rate limits.
Cancel disabled	No manual analysis is currently running.
Success counter resets	It is session-level; persisted per-source Last refresh remains available.
Expected candle differs	Check market holiday or data-provider availability; holiday detection is not automatic.

15 Review and validation worksheet

Use several known valid and invalid symbols for each profile. Compare chart values with the row audit before judging alert quality.

Check	Expected evidence	Reviewer
Physical touch	low <= captured Purple <= high; close from 0% through +3%	Pass / Fail
Strict setup	Black, EMA26, blue/Purple, approach, confirmation, structure	Pass / Fail
Persistence	Same touch updates same setup ID; new touch creates new lifecycle	Pass / Fail
Fixed threshold	Captured EMA9 x 1.03 does not move; equality valid	Pass / Fail
C1/C2	Both eligible; C3+ uses lower C1/C2 low forever	Pass / Fail
Early	Closed entry candle close strictly above yellow	Pass / Fail
Final	Close > yellow and yellow > brown	Pass / Fail
Exit	Only corresponding open trade closes when exit candle close < yellow	Pass / Fail
Freshness	Stale/missing data creates no entry, exit, or Telegram alert	Pass / Fail
Concurrency	Profiles, Entry, and Exit progress independently; downloads stay rate limited	Pass / Fail

Implementation confidence

Automated tests cover persistence, fixed 3%, C1/C2 entries and invalidation, Early-to-Final precedence, open-trade exits, stale-data blocking, duplicate candle suppression, profile mappings, scheduling priorities, concurrent analysis, cancellation, and UI/API payloads. Chart review remains essential for strategy validation.